

Physical Characterization of Greenhouse Substrates for Automated Irrigation Management

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Keywords: greenhouse substrates, physical properties, water characteristic curve

Abstract

Over the last decade, the greenhouse industry experienced a significant increase in production capacity in response to enhanced demand of high-quality crops. To optimize yield and quality of greenhouse crops, substrates with optimal balance of aeration and water holding properties are essential. A wide variety of root zone media such as perlite, rockwool, coco coir, foamed glass or mixtures of these substrates have been successfully used in greenhouse agriculture. There is mounting empirical evidence that dual porosity (i.e., aggregated) media that contain small intra-aggregate pores for water storage and larger inter-aggregate pores for aeration create an improved rhizosphere environment for many crops. To investigate the suitability of these substrates and various mixtures thereof for cultivation of tomatoes, we conducted a comprehensive measurement campaign to characterize water retention properties. The measured substrate water retention curves exhibit various unimodal (e.g., coco coir and rockwool) and bimodal shapes (e.g., foamed glass and perlite) with differing air entry potentials, providing valuable information for irrigation scheduling to balance water storage and aeration for optimum growth conditions. Furthermore, the obtained curves can be used to parameterize numerical simulation models for optimization of irrigation strategies and other controllable environmental variables.

INTRODUCTION

In the arid Southwest of the United States, greenhouse agriculture must compete for scarce water resources. To keep production costs within reasonable limits and to conserve water, smart irrigation systems adjusted to particular growth substrates and crops are essential. Irrigation frequency and duration directly affect availability of water and oxygen in the root zone, which have to be in balance to obtain optimum growth and yield. A common method to schedule irrigation within soilless culture production is based on solar radiation. Dorais et al. (2005) show that about 1 mm of water per 1 J cm^{-2} measured radiation is sufficient for tomatoes in sawdust-based and rockwool substrates. Another method used by greenhouse growers is to irrigate based on the amount of drainage. The leaching fraction, defined as the ratio of drainage and applied irrigation water, differs for various crops, water quality and substrates. It helps to prevent salt accumulation and also provides fresh oxygen to the root zone. For tomatoes grown on rockwool, the target leaching fraction is typically between 10–40% per day (Peet, 2005). Other irrigation scheduling methods rely on either measuring the volumetric water content or the matric potential. Electromagnetic techniques such as Time Domain Reflectometry (TDR), Capacitance and Frequency Domain methods (e.g., ECH₂O sensors), or Amplitude Domain Reflectometry (e.g., Theta impedance sensors) are commonly applied to measure volumetric water content (Or et al., 2006). These methods capitalize on the large disparity between the dielectric constants of water (~81), substrate (3–5) and air (~1), and employ empirical or physically-based relationships to relate the measured bulk dielectric constant or permittivity to water content. Nemali et al. (2007)

developed calibration relationships for ECH₂O sensors (Decagon Devices Inc.) and Theta probes (Delta-T Devices Ltd.) for various organic (peat and pine bark) and inorganic substrates (perlite and vermiculite) and mixtures thereof.

Electromagnetic methods work well for homogenous substrates with small particle sizes (e.g., rockwool or coco coir), yet they tend to fail in aggregated and structured media (e.g., perlite or foamed glass) due to lack of intimate contact between sensors and substrate. Another shortcoming is that most sensors require time consuming, substrate specific calibration to convert bulk dielectric constant to water content. To avoid these shortcomings, tensiometers can be employed to measure the substrate matric potential and relate it to water content via substrate water retention relationships. The substrate matric potential can be also used as direct indicator for forthcoming stress situations and as trigger to initiate and cease irrigation. Klein (2004) used a CommonSensor™ tensiometer developed for potted plants in soilless media for successful automatic irrigation. To optimize irrigation and available plant water content based on matric potential measurements and to a priori simulate water and gas distribution within various substrates, a continuous potential – water content relationship (retention curve) needs to be established via measurements and parametric models.

The presented study is a first step towards the design of smart irrigation strategies based on continuous monitoring of substrate matric potential. This requires comprehensive characterization of substrate retention properties as demonstrated in the following sections.

MATERIALS AND METHODS

To cover a wide range of commonly employed growth substrates, we determined retention functions for: (a) coco coir (RIOCOCO, Sri Lanka); (b) rockwool (Expert™, Grodan®, The Netherlands); (c) horticultural perlite (Therm-O-Rock West Inc., USA); (d) foamed glass with aggregate diameters ≤1.27 cm (Growstones™, Earthstone LLC., USA); and (e) a 50/50 mix (volume basis) of coco coir and foamed glass aggregates (1.27–1.91 cm average diameter). Considered media are shown in Figure 1 with relevant properties listed in Table 1.

Two different methods, pressure desaturation and a chilled-mirror dew point technique (Gee et al., 1992) were used for high and low matric potential ranges, respectively. For the high potential range (-0.5 to -200 cm) we employed Tempe cells (Soilmoisture Equipment Corp., USA) connected to a manifold with high resolution pressure regulator and high-precision, digital pressure gauge. The stepwise applied pressure was additionally monitored with water and mercury U-tube manometers. Prior to the experiment all samples were slowly saturated from the bottom to minimize potential air entrapment. After determining the mass of the empty cells and the mass of the cells including the saturated samples, pressures of 0.5, 1.5, 3, 5, 10, 20, 30, 50, 75 and 100 cm, respectively were sequentially applied. Each pressure step was maintained until the sample was in equilibrium and outflow ceased. After each pressure step, samples were disconnected from the manifold and weighed to determine the associated volumetric water content from wet and dry sample mass (the dry mass was determined after the last pressure step via oven drying at 105°C for 48 hours. For a detailed description of the pressure desaturation method, see Tuller and Or (2004).

For the low matric potential range (-7000 to -10⁶ cm), a temperature-compensated WP4-T Dewpoint Meter (Decagon Devices Inc., USA) was used. The WP4-T operates based on the chilled-mirror dew point technique (Gee et al., 1992) and enables rapid determination of substrate water potential in equilibrium with water vapor in a sealed chamber. At equilibrium, the relative humidity in the chamber at a given temperature was used to determine the sample water potential. A detailed description of this method and associated sample preparation procedure is provided in Tuller and Or (2005).

After all water content – matric potential data pairs were measured, the water contents were normalized by the respective saturated water content to obtain relative saturation (S_e). Nonlinear regression analysis (least square fit method) was applied to

parameterize two empirical retention models. The most commonly applied retention model valid for homogeneous, fine textured substrates was developed by van Genuchten (1980) and is given as:

$$S_e = \left[\frac{1}{1 + (\alpha |\psi_m|)^n} \right]^m \quad (1)$$

where, S_e is the effective saturation, ψ_m the matric potential, and α , n and m are empirical parameters determined by the shape of the $S_e(\psi_m)$ curve. This model was extended by Durner (1994) to calculate retention curves for aggregated media with two distinct pore spaces (intra-aggregate and inter-aggregate pores) such as perlite or foamed glass. The extended model is based on superposition of two van Genuchten-type subcurves:

$$S_e = w_1 \left[\frac{1}{1 + (\alpha_1 |\psi_m|)^{n_1}} \right]^{m_1} + w_2 \left[\frac{1}{1 + (\alpha_2 |\psi_m|)^{n_2}} \right]^{m_2} \quad (2)$$

where, w_1 and w_2 are weighing factors that need to sum up to unity. Both models are commonly parameterized for a particular substrate by matching eqn. 1 or eqn. 2 with measured data via nonlinear regression analysis.

RESULTS AND DISCUSSION

The results are presented in Figure 2 with inlays listing obtained van Genuchten (1980) (VG) and Durner (1994) (VG-D) model parameters. The VG model was successfully applied to coco coir (Fig. 1a) and rockwool (Fig. 1b), substrates with unimodal pore size distributions. Both, coco coir and rockwool have air entry values of about -1 to -2 cm. The air entry value is the potential at which the largest pores in the system start to drain and open up pathways for aeration. After air entry, the retention curve for rockwool exhibits a much gentler slope than the curve for coco coir. This indicates that in rockwool a major portion of the larger pores drains almost simultaneously, which promotes aeration, but also significantly reduces plant available water content in the potential range of interest for plant growth. This behavior needs to be considered for development of smart irrigation strategies that are based on continuously monitored substrate water status. All other investigated substrates (perlite, foamed glass aggregates, and coco coir-foamed glass aggregate mix) (Figs. 1c-e) have bimodal pore size distributions with two distinct groups of pores. The first group comprises of large inter-aggregate (structural pores) that drain almost simultaneously at relatively high potentials. These pores are preferred pathways for substrate aeration and allow oxygen to quickly diffuse at air water interfaces. The second group is comprised of small intra-aggregate pores that are utilized as storage compartments for plant water supply. Again, the location of the two air entry points and the overall shape of the retention curve can be used to optimize irrigation amount and frequency. Based on the shape of the retention curve it is obvious that perlite and the coco coir – foamed glass mix require less frequent irrigation due to the higher water storage capacity of the intra-aggregate pore domain.

CONCLUSIONS

Data such as presented in this manuscript are crucial for triggering start and cessation of irrigation systems that are based on monitoring of substrate water status. The parameterized continuous VG and VG-D functions (Eqns. 1 and 2) can be used as input parameters for numerical simulation models to optimize irrigation schedules for water supply and aeration, which is one of the next steps in our study. Simulation results provide valuable information to researches prior to field trails and aid with selection of favorable growth media – irrigation strategy combinations. This allows better informed research and management and helps avoid costly and time consuming mistrials.

ACKNOWLEDGMENTS

Research support provided by Earthstone, LLC, the University of Arizona Center for Environmental Physics and Mineralogy (CEPM), and the Controlled Environment Agriculture Center (CEAC), The University of Arizona. CEAC Publication P-125933-14-08.

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Tables

Table 1. Physical properties of investigated growth substrates.

Substrate	Aggregate/particle size (cm)	Bulk density (g/cm ³)	Air porosity
Coco Coir	≤1.91	0.061	0.90
Rockwool	----	0.050	0.97
Perlite	≤0.50	0.131	0.55
Foamed Glass Aggregates	≤1.27	0.152	0.71
Coco Coir - Foamed Glass Mix	1.91–1.27	0.119	0.83

Figures

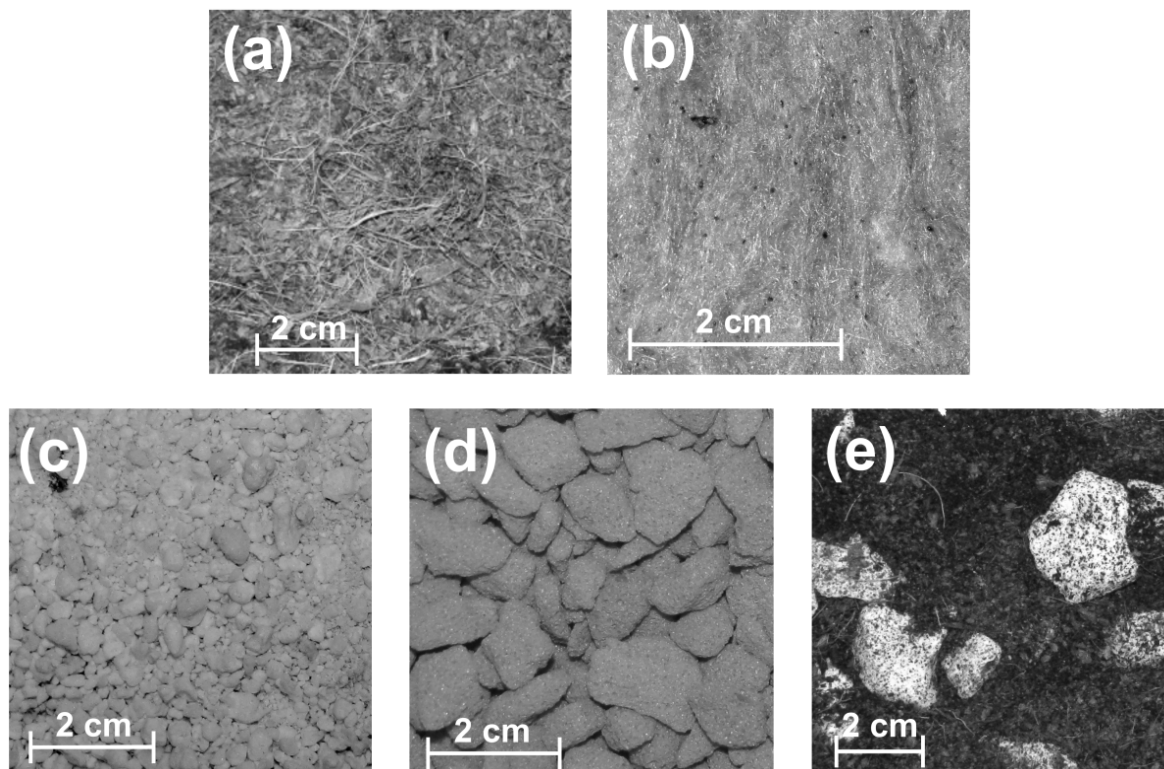


Fig. 1. Investigated growth substrates: (a) coco coir; (b) rockwool; (c) perlite; (d) foamed glass aggregates; and (e) 50/50 coco coir – foamed glass aggregate mix.

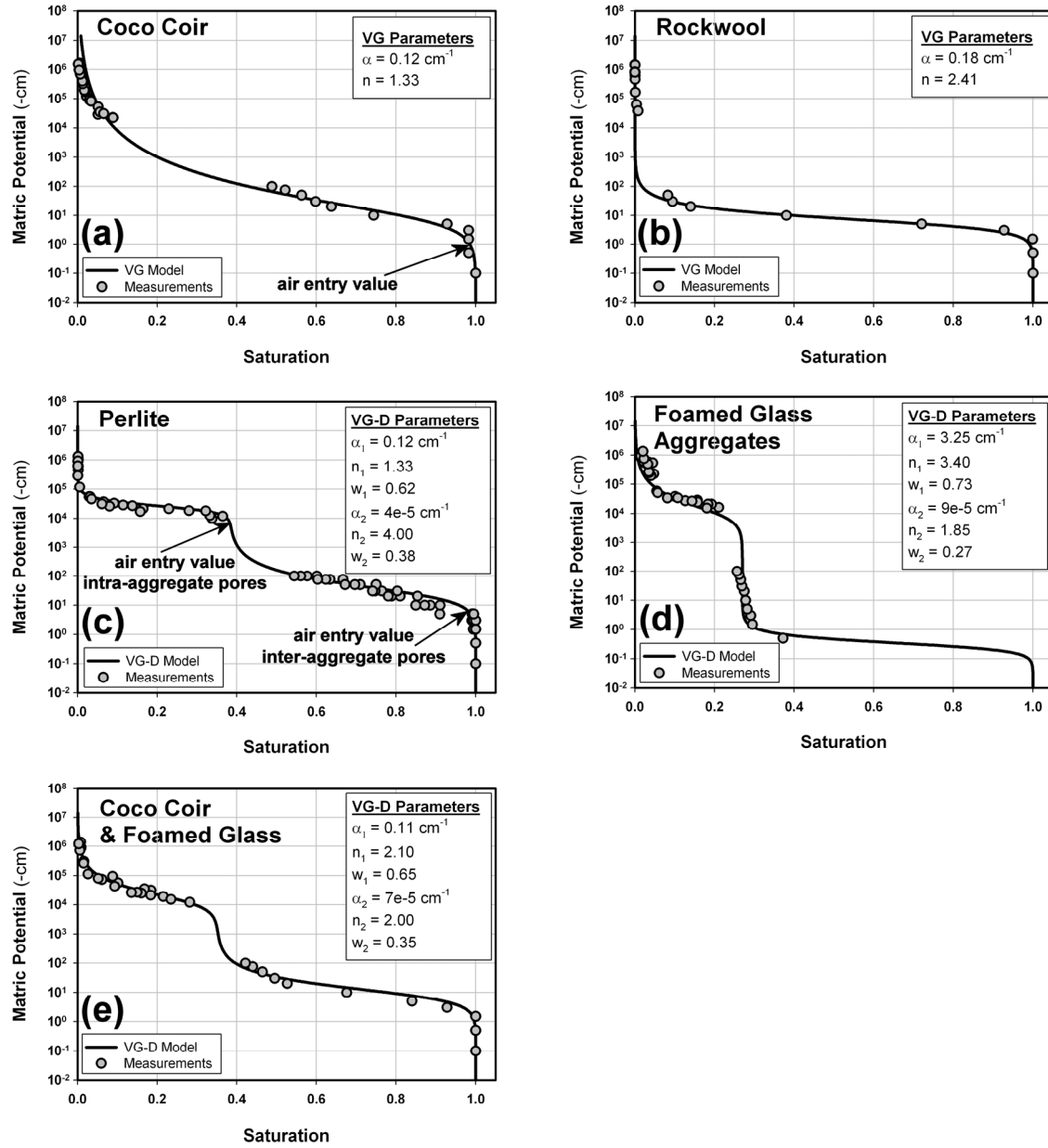


Fig. 2. Measured and modeled retention curves for: (a) coco coir; (b) rockwool; (c) perlite; (d) foamed glass aggregates; and (e) 50/50 coco coir – foamed glass aggregate mix (inlays list the VG or VG-D model parameters).